

YEAR 7 • UNIT 1 • HOMEWORK 2

Signs, Multiples and Cells

Mathematics 1.2 — Multiplying & Dividing Integers • 1.3 — Multiples and the LCM
 Science 1.2 — Plant & Animal Cells • 1.3 — Specialised Cells

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**.
- In science, answer in **full English sentences**. “Haemoglobin” is not a sentence; “A red blood cell is full of haemoglobin” is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A • Mathematics 1.2 — Multiplying and Dividing Integers

Remember from the lesson

Same signs give a positive. Different signs give a negative. The same four rules work for $\times$ and for $\div$, because dividing undoes multiplying.

A **product** is the answer when you multiply. The product of 2 and -9 is -18 .

A1 The four rules

(a) Fill in the sign that each rule gives you.

$$+ \times + = \underline{\quad} \quad + \times - = \underline{\quad} \quad - \times + = \underline{\quad} \quad - \times - = \underline{\quad}$$

(b) Now work these out.

$$(a) -6 \times 4 = \underline{\quad} \quad (d) -36 \div 6 = \underline{\quad} \quad (g) -9 \times 0 = \underline{\quad}$$

$$(b) 7 \times -3 = \underline{\quad} \quad (e) 45 \div -9 = \underline{\quad} \quad (h) -12 \div -12 = \underline{\quad}$$

$$(c) -5 \times -8 = \underline{\quad} \quad (f) -56 \div -7 = \underline{\quad} \quad (i) 8 \times -7 = \underline{\quad}$$

(j) $-3 \times -3 =$ _____

(k) $-100 \div 4 =$ _____

(l) $-60 \div -5 =$ _____

A2 Say it, then write it**Word help – the English decides which number comes first****multiply -7 by 6** you **start** at -7 : -7×6 .**divide -40 by 8** you start at -40 : $-40 \div 8$.**5 divided into -35** the same trap as last week – it is $-35 \div 5$, not $5 \div -35$.**the product of**the answer when you **multiply**.**times**

the everyday English word for multiply.

share equally between

divide.

Turn each sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Multiply
- -7
- by
- 6
- .

Calculation: _____ Answer: _____

2. Divide
- -40
- by
- 8
- .

Calculation: _____ Answer: _____

3. What is the product of
- -9
- and
- -4
- ?

Calculation: _____ Answer: _____

4. 5 divided into
- -35
- .

Calculation: _____ Answer: _____

5. Mr Bowen shares a debt of 24 dollars equally between 6 people. Write each person's share as a negative number.

Calculation: _____ Answer: _____

A3 A sentence that is only half true**Think carefully about this one**People often say: **"Two negatives make a positive."**It is a useful sentence, but it is only true *sometimes*. Test it yourself before you decide.(a) Work out -3×-4 Answer: _____(b) Work out $-3 + -4$ Answer: _____(c) Is "two negatives make a positive" true for **both** of them? Answer in one full sentence.

(d) Rewrite the sentence so that it is **always** true. You will need to say which operations it works for.

A4 Brackets, and checking yourself

Two rules that save marks

Do the brackets first. Work out what is inside, then multiply or divide. Careful: if the bracket holds an **addition**, the sign rules above do **not** apply inside it.

Estimate: round each number to something easy, calculate, and check your real answer is close to it.

(a) Work these out. Show what the bracket comes to first.

(i) $20 \div (-3 + -2) =$ _____

(ii) $(-4 + 10) \times -3 =$ _____

(iii) $-6 \times (2 - 9) =$ _____

(b) **Estimate** each of these by rounding first. Write the rounded calculation as well as the answer.

(iv) -6.2×3.9 is about _____ = _____

(v) $62 \div -7.8$ is about _____ = _____

A5 Word problems

Read each one **all the way to the end** before you start writing.

1. **The monthly fee.** Mr Bowen's bank takes a 4-dollar fee out of his account at the end of every month. He forgets about it for 7 months. What is the total change to his balance?

2. **The dive.** A submarine starts at the surface and descends steadily. After 6 minutes it is at -54 m. How far does it travel each minute?

3. **The market stall.** Mr Bowen takes 47 identical rubber ducks to the market to sell. He plans to sell them for 3 dollars each, and the market charges him a 2-dollar fee on every duck he sells, plus 15 dollars for the stall. It rains all day and nobody comes, so he sells 0 ducks. How much money does he make or lose that day?

Section B • Mathematics 1.3 – Multiples and the LCM

Remember from the lesson

A **multiple** is the answer when you multiply a number by 1, 2, 3, 4, ... The multiples of 4 are 4, 8, 12, 16, 20, ...

In everyday English, **common** means *ordinary*. In maths it means **shared**. A **common multiple** of two numbers is in **both** lists, and the **lowest common multiple (LCM)** is the smallest of them. You already know this idea in Vietnamese as **BCNN**.

B1 Build it from the lists

- (a) Write the first six multiples of 8.

- (b) Write the first six multiples of 12.

- (c) Which numbers appear in **both** lists? These are the common multiples.

- (d) So the **LCM** of 8 and 12 is _____

B2 Do not just multiply them

The trap in this topic

The LCM is usually **smaller** than the two numbers multiplied together. If one number **divides into** the other, the LCM is just the **bigger** number. When you are not sure, **list them**.

Find the LCM of each pair.

(a) 3 and 7 LCM = _____

(c) 6 and 8 LCM = _____

(b) 5 and 20 LCM = _____

(d) 9 and 9 LCM = _____

(e) A student says the LCM of 5 and 20 is 100, because $5 \times 20 = 100$. In one full sentence, explain what they have got wrong.

B3 Word problems

1. **The two lighthouses.** One lighthouse flashes every 6 seconds. The other flashes every 10 seconds. They have just flashed at the same moment. How many seconds until they next flash together?

2. **The bread and the cheese.** Mr Bowen is making sandwiches. Bread rolls come in packs of 8. Slices of cheese come in packs of 12. He wants one slice of cheese in every roll, with nothing left over. What is the smallest number of sandwiches he can make, and how many packs of each must he buy?

B4 Now you write the question

Your turn to be the teacher

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths – write the English.

Your word problem must:

- be about something **real**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

(a) Write a word problem for

$$7 \times -4 = -28$$

(b) Write a word problem for

$$\text{LCM of 4 and 10} = 20$$

This one must be about two things that keep repeating, and the question must ask when they next happen together.

Section C • Science 1.2 – Comparing Plant and Animal Cells**Remember from the lesson**

An animal cell has a **cell membrane**, **cytoplasm**, **mitochondria** and a **nucleus**. A plant cell has all four of those **plus** a **cell wall**, **chloroplasts** and a **sap vacuole**.

The one clue that always works is the **cell wall** – a straight, stiff edge, with the cells packed together like bricks. **“Not green” does not mean animal.**

C1 Match the word to its meaning

Write the correct letter in the box next to each word.

Word	Letter	Meaning
1. Stain		A The job that a cell does.
2. Similar		B The tough substance a cell wall is made of.
3. Function		C A coloured dye added to a specimen to make its parts easier to see.
4. Specialised		D The green substance inside a chloroplast.
5. Cellulose		E Alike in some ways and different in others – not the same.
6. Chlorophyll		F The cell has a structure that helps it do its job really well.

C2 Plant or animal? And how do you know?

Mr Bowen looks at four slides down the microscope. For each one, write **plant** or **animal**, and then give the **one feature** that tells you.

	What Mr Bowen sees	Plant or animal?	The feature that tells you
1	Cells with straight, stiff edges, packed together like bricks, and full of small green discs.		
2	Round, separate cells with no straight edges anywhere.		
3	Purple cells with no green discs at all – but every cell has straight sides.		
4	Long, soft fibres with many dark nuclei, and no straight edges.		

One of these four catches almost everybody. Read the orange box at the top of this section again before you answer.

C3 Similar is not the same

Word help – three patterns for comparing

“A and B are **similar because** both of them ...”

“They are **different because** A has ..., **but** B ...”

“**Unlike** an animal cell, a plant cell ...”

1. Finish this sentence: “A plant cell and an animal cell are **similar because** both of them ...”

2. Finish this one: “They are **different because** a plant cell has ..., **but** an animal cell ...”

3. An animal cell has no fixed shape. Explain why, in one full sentence.

C4 Put the method in order

These are the steps for making a stained slide of your own cheek cells, but they are in the wrong order. Number them **1** to **6**.

☐	Add one drop of methylene blue stain to the sample.
☐	Looking from the side , move the lens down close to the slide.
☐	Rub the inside of your cheek gently with a clean cotton bud.
☐	Look down the eyepiece and turn the focus so the lens moves away , until the cells are sharp.
☐	Lower a cover slip onto the drop slowly, so that no air bubbles are trapped.
☐	Wipe the cotton bud onto the middle of a clean glass slide.

Why must you look **from the side** in one of those steps? Answer in one full sentence.

Section D • Science 1.3 – Specialised Cells

Remember from the lesson

A cell's **function** is the job it does. A **specialised** or **adapted** cell has a structure that helps it do that job really well.

The sentence to use every time: A [cell] is **adapted to** [its job] **because it has** [its special feature].

D1 Finish the table

The first row is done for you. Fill in the other four.

Name	Function (its job)	Special structure	How this helps
Red blood cell	Carries oxygen around the body.	Full of haemoglobin, and no nucleus.	Haemoglobin holds the oxygen, and losing the nucleus leaves more room for it.
Neurone			
Ciliated cell			
Root hair cell			
Palisade cell			

D2 Use the sentence, and think

1. Write the full sentence for the **root hair cell**, using **adapted to** and **because it has**.

2. Write the full sentence for the **palisade cell**.

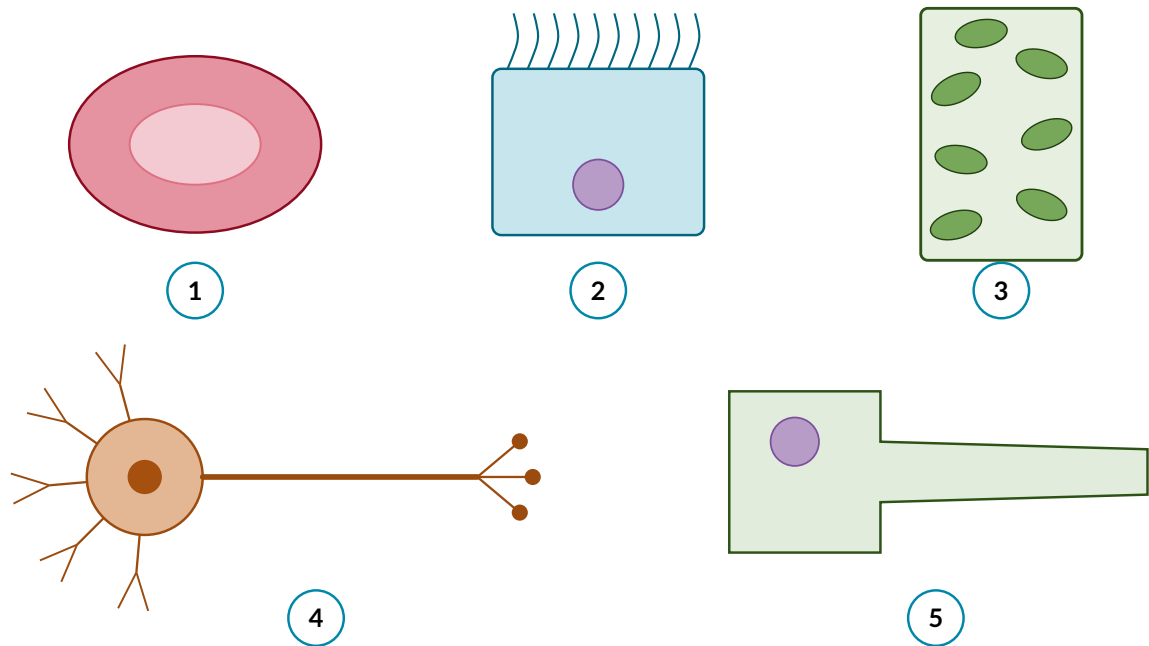
3. A red blood cell, a neurone and a ciliated cell are all animal cells. Name **two** things that all three of them have.

Careful. Check each one before you write.

4. Root hair cells have **no** chloroplasts. Suggest why not.

D3 Name the cell

Each drawing is one of the five specialised cells. Write its name and its job in the table below.



No.	Name of the cell	Its job
1		
2		
3		
4		
5		

Answer Key — Teacher Copy

A1 The four rules

(a) +, −, −, +. Same signs positive, different signs negative.

(b) (a) −24 (b) −21 (c) 40 (d) −6 (e) −5 (f) 8

(g) 0 (h) 1 (i) −56 (j) 9 (k) −25 (l) 12

(g) catches people who reach for a sign rule: zero has no sign, and anything times zero is zero. (h) is worth a comment too — a negative divided by itself is 1, not −1.

A2 Say it, then write it

1. $-7 \times 6 = -42$

2. $-40 \div 8 = -5$

3. $-9 \times -4 = 36$

4. $-35 \div 5 = -7$

5. $-24 \div 6 = -4$ dollars each

Question 4 is the reversed-order trap carried over from Homework 1: “5 divided into −35” is $-35 \div 5$. A student who writes $5 \div -35$ has made the English error, not an arithmetic one — make them re-read the sentence rather than re-do the sum.

A3 A sentence that is only half true

(a) 12 (b) −7

(c) No. It is true for -3×-4 but false for $-3 + -4$, which is still negative.

(d) Anything that attaches the operation, e.g. “Two negatives make a positive when you *multiply* or *divide*, but not when you add.”

This is the whole point of Maths 1.2 and the question most worth marking properly. Accept any wording that names multiplying and dividing and excludes adding; reject a (d) that just repeats the original sentence more loudly. If most of the class gets (c) right and (d) vague, that is a five-minute discussion, not a pile of red pen.

A4 Brackets, and checking yourself

(i) $-3 + -2 = -5$, so $20 \div -5 = -4$

(ii) $-4 + 10 = 6$, so $6 \times -3 = -18$

(iii) $2 - 9 = -7$, so $-6 \times -7 = 42$

(iv) about $-6 \times 4 = -24$

(v) about $64 \div -8 = -8$

The classic error in (i) is applying the new sign rule inside the bracket and getting +5. The bracket holds an *addition*: two negatives added stay negative. Watch for it specifically.

A5 Word problems

1. $7 \times -4 = -28$. His balance goes down by 28 dollars.

2. $-54 \div 6 = -9$, so 9 m downwards each minute.

3. He loses **15 dollars**. He sells 0 ducks, so $0 \times 3 = 0$ income and $0 \times -2 = 0$ in selling fees — but the 15-dollar stall fee is charged anyway. The 47 and the 3 and the 2 are all decoration.

Problem 3 is the deadpan one and should be read completely straight. It is a harder trick than last week's: most of the numbers cancel to zero, but not *all* of them, so a student who answers "0" has stopped one step early. Expect both "0" and "−15" in the room and make them argue it out.

B1 Build it from the lists

- (a) 8, 16, 24, 32, 40, 48
- (b) 12, 24, 36, 48, 60, 72
- (c) 24 and 48
- (d) 24

B2 Do not just multiply them

- (a) 21 (b) 20 (c) 24 (d) 9
- (e) 5 divides into 20, so the LCM is just the bigger number, 20. Multiplying only gives the right answer when the two numbers have no common factor except 1 — as in (a), where $3 \times 7 = 21$ happens to be correct.
- (d) is the deadpan one: both numbers are 9, so there is nothing to work out. Expect someone to list the multiples of 9 twice before noticing.

B3 Word problems

1. $\text{LCM}(6, 10) = 30$, so 30 seconds. Not 60 — 6 and 10 share the factor 2.
2. $\text{LCM}(8, 12) = 24$. So 24 sandwiches: **3 packs of rolls** and **2 packs of cheese**. Students who stop at 24 have done the maths but not answered the question; the packs are the point.

B4 Now you write the question

No single right answer. Check, in this order:

1. **Does it repeat?** (a) needs something happening 7 times, each time costing 4 — a fee, a drop in temperature, a debt. A one-off "he lost 28 dollars" is not a multiplication.
2. **Is the sign right?** The −4 must be a loss, a fall or a descent, not a gain.
3. **For (b), are the two things independent and repeating?** Buses, lights, alarms, taps, machines. The question must ask when they *next coincide*, not how many of each.
4. **Full sentences, ending in a question?**
5. **Does their own answer come out as −28, and as 20?**

One that works for (b): "One machine in a factory stamps a box every 4 seconds. Another wraps a box every 10 seconds. They have just both finished a box at the same moment. How many seconds until they finish together again?"

Expect (b) to be much weaker than (a) — writing a repeating-events story is harder than writing a repeated-loss one. Worth doing one aloud on the board.

C1 Match the word to its meaning

1-C 2-E 3-A 4-F 5-B 6-D

C2 Plant or animal?

1. **Plant** — straight, stiff cell walls (and chloroplasts).
2. **Animal** — no cell wall, so no straight edges.
3. **Plant** — every cell has a straight-sided *cell wall*. This is the trap. It is Rhoeo leaf epidermis: purple, with no chloroplasts at all, because it is not a photosynthesising layer.

4. **Animal** — no cell walls; these are muscle fibres.

Number 3 catches almost everybody, because they have quietly replaced “has a cell wall” with “is green”. Do not mark it as a careless slip — it is the misconception the whole lesson was built to break. Ask them what the onion looked like: also a plant, also no chloroplasts.

C3 Similar is not the same

1. ...both of them have a cell membrane, cytoplasm, a nucleus and mitochondria. (Any of those four.)
2. ...a plant cell has a cell wall (or chloroplasts, or a sap vacuole), but an animal cell does not.
3. An animal cell has no cell wall, so there is nothing stiff to hold it in a fixed shape — only the thin, flexible cell membrane.

The word to police here is *similar*. A student who writes “they are the same” has lost the distinction the lesson taught, even if the rest of the sentence is right.

C4 Put the method in order

Reading down the table as printed: **3, 5, 1, 6, 4, 2.**

In order: (1) rub the inside of your cheek with a cotton bud; (2) wipe it onto the middle of a clean slide; (3) add a drop of methylene blue stain; (4) lower a cover slip slowly to avoid bubbles; (5) looking from the side, move the lens down close to the slide; (6) look down the eyepiece and focus by moving the lens away.

Why from the side: with your eye at the eyepiece you cannot judge the gap, so looking from the side is the only way to avoid driving the lens through the slide.

Focusing *away* from the slide, never towards it, is the same safety point said twice. Accept either the safety answer or “so you do not break the slide or the lens”.

D1 Finish the table

Neurone	carries electrical signals	a very long axon; short dendrites	signals travel far and fast; dendrites collect them
Ciliated	sweeps mucus away from the lungs	tiny moving hairs (cilia)	they sweep the mucus, with its trapped dust and germs, along
Root hair	absorbs water from the soil	a long, thin extension	a big surface area, so water moves in easily
Palisade	makes food by photosynthesis	packed with chloroplasts, near the top of the leaf	catches as much sunlight as possible

D2 Use the sentence, and think

1. “A root hair cell is adapted to absorb water because it has a long, thin extension that gives it a big surface area.”
2. “A palisade cell is adapted to make food by photosynthesis because it has many chloroplasts.”
3. **Cytoplasm** and a **cell membrane**. Not a nucleus — the red blood cell has none. Accept “they are all specialised” as a third.
4. Roots are underground in the dark, and chloroplasts need sunlight to make food, so they would be useless there.

Question 3 is the precision point of Science 1.3, and nearly everyone writes “a nucleus”. Let them, then ask what a red blood cell gave up to fit more haemoglobin in. In 1 and 2, mark the *frame* as much as the content: if “adapted to” and “because it has” are both missing, the science may be right but the sentence is not the one being taught.

D3 Name the cell

- 1 — red blood cell (carries oxygen)
- 2 — ciliated cell (sweeps mucus away from the lungs)
- 3 — palisade cell (makes food by photosynthesis)
- 4 — neurone (carries electrical signals)
- 5 — root hair cell (absorbs water from the soil)

2 and 5 are the pair most often swapped: both are drawn as a box with something sticking out of it. The cilia are *many* short hairs on the top; the root hair is *one* long extension from the side.

If anyone labels the pale middle of 1 as a nucleus, that is the misconception from D2.3 showing up again — a red blood cell has no nucleus, and the pale patch is the dip in the disc.